

ABDULLA AL HUSSAIN

Software Engineer | Full-Stack Developer | AI Researcher

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PROFESSIONAL SUMMARY

Results-driven B.Sc. CSE graduate (CGPA 3.78/4.00, Varendra University, 2026) with hands-on expertise in full-stack web development, machine learning, and academic research. Proficient in building scalable, production-ready applications with React, Node.js, MongoDB, and Firebase. IEEE-published researcher in deep learning applied to medical image analysis. Demonstrated ability to deliver end-to-end software solutions, collaborate in teams, and communicate complex technical concepts clearly. Seeking a challenging software engineering, web development, or AI/ML role to contribute technical skills and drive impactful outcomes.

EDUCATION

B.Sc. in Computer Science & Engineering

2022 – 2026

Varendra University, Rajshahi, Bangladesh | UGC-Approved

- CGPA: 3.78 / 4.00
- Key coursework: Data Structures & Algorithms, Machine Learning, Database Management Systems, Software Engineering, Computer Networks, OOP, Operating Systems

Higher Secondary Certificate (HSC) – Science

2020

Shahid Buddhijibi Government College, Rajshahi

- GPA: 5.00 / 5.00 — Perfect score

Secondary School Certificate (SSC) – Science

2018

River View Collectorate School, Rajshahi

- GPA: 5.00 / 5.00 — Perfect score

WORK EXPERIENCE

Research Assistant – Financial Data Analysis

2026

Collaborative Research Project | Massey University MBA Research

- Conducted end-to-end data analysis for a study on the causal impact of inflation on stock market returns using advanced econometric methods: Propensity Score Matching (PSM), Inverse Probability of Treatment Weighting (IPTW), and Doubly Robust (DR) estimation.
- Built data pipelines in Python to preprocess, clean, and model large financial datasets, producing findings that confirmed a statistically significant negative relationship between inflation and stock returns.
- Authored technical documentation and research reports, demonstrating strong written communication and analytical rigor.

RESEARCH & PUBLICATIONS

IEEE ICCIT 2025 – Published Research Paper

2025

IEEE Xplore | DOI: 10.1109/ICCIT68739.2025.11491822

- Designed and implemented a deep transfer learning pipeline for the detection and multi-class severity classification of diabetic retinopathy from retinal fundus images.
- Fine-tuned pre-trained CNN architectures (transfer learning) to achieve high classification accuracy on medical imaging datasets, contributing toward accessible AI-assisted ophthalmology diagnostics.
- Managed complete research lifecycle: literature review, model development, training, evaluation, and peer-reviewed publication in IEEE Xplore.

PROJECTS

Hope's Rising – Full-Stack Donation Platform

2024

React · Node.js · Firebase · REST API

- Architected and deployed a real-time, multi-category donation platform with secure financial transaction handling, Firebase authentication, and role-based access control.
- Implemented responsive UI with real-time data sync via Firebase Firestore, reducing data latency and supporting concurrent users across multiple donation categories.
- Integrated secure payment flows and user verification to ensure data integrity and platform trustworthiness.

Book Store – Full-Stack E-Commerce Application

2023 – 2024

React · Node.js · MongoDB · Express.js

- Developed a complete e-commerce web application with product listing, dynamic category filtering, cart management, and secure JWT-based user authentication.
- Designed RESTful API endpoints and MongoDB schemas for efficient data retrieval, enabling scalable product and order management.

Notepad – Real-Time Note Management App

2023

React · Node.js · MongoDB

- Built a full-stack CRUD application allowing users to create, edit, organize, and search notes in real time with persistent cloud storage and instant UI updates.

Hydro Harmony – Hult Prize Innovation Project

2024

Global Social Entrepreneurship Competition

- Led the technical development of a safe water accessibility solution for the global Hult Prize competition; presented the scalable model to an international panel of evaluators.
- Demonstrated product thinking, cross-functional collaboration, and the ability to translate social impact goals into a technically feasible prototype.

HTML & CSS / Front-End Projects

2023 – 2024

HTML5 · CSS3 · Tailwind CSS · DaisyUI | Live on GitHub

- Built and published multiple responsive front-end projects showcasing layout design, component styling, and UI best practices using Tailwind CSS and DaisyUI.

TECHNICAL SKILLS

Languages:	Python, C, C++, Java, JavaScript (ES6+)
Frontend:	React.js, HTML5, CSS3, Tailwind CSS, DaisyUI, Responsive Design
Backend:	Node.js, Express.js, REST API Design
Databases:	MongoDB, Firebase Firestore, Firebase Realtime Database
AI / ML:	Deep Learning, Transfer Learning, CNNs, TensorFlow, Keras, Scikit-learn
Data / Stats:	Python (Pandas, NumPy), Econometric Modeling (PSM, IPTW, DR Estimation)
Dev Tools:	Git, GitHub, VS Code, Postman, npm
Other:	MS Office Suite (Word, Excel, PowerPoint), Technical Writing, Agile Basics

ACHIEVEMENTS & RECOGNITION

- Published research paper in IEEE Xplore (ICCIT 2025) — peer-reviewed international conference publication.
- Hult Prize 2024 Participant — one of the world's largest student social entrepreneurship competitions; represented Varendra University.
- HSC & SSC Perfect GPA (5.00/5.00) — consistent academic excellence across secondary education.
- Runners-up, Intra-Department Football Tournament — demonstrates leadership, teamwork, and competitive performance.

LANGUAGES

English: Fluent — technical writing, documentation, professional communication, presentations

Bengali: Native proficiency — verbal and written

CORE COMPETENCIES

Analytical Thinking · Problem-Solving · Time Management · Team Collaboration · Presentation Skills · Networking · Negotiation · Attention to Detail · Self-Motivated · Fast Learner